

2018-Ph-H1-Q13

Section: Electricity

Topic: Internal Resistance

A 6.0 V supply with internal resistance 1.0 Ω is connected to a 5.0 Ω resistor.

What is the voltage across the 5.0 Ω resistor?

Solution:

Total resistance = 5.0 + 1.0 = 6.0 Ω

Total current:

$$I = \frac{E}{R_{total}} = \frac{6.0}{6.0} = 1.0 \text{ A}$$

Voltage across 5.0 Ω resistor:

$$V = IR = 1.0 \times 5.0 = 5.0 \text{ V}$$

Answer: 5.0 V

Revision Tip:

Use internal + external resistance to get total current, then apply $V = IR$ to find the external voltage.